
TO: ANYONE
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Free
Transcription
Stack
XEROX
TELECOPIER

@dandjlab

THE FREE STACK

01	**ffmpeg**	clean the audio `mono 16khz wav`
02	**whisper**	the words and the timing `faster-whisper 1.2.1 / large-v3 pinned`
03	**pyannote**	who speaks when `pyannote.audio 4.0.7 / community-1`
04	**numpy**	pause, energy, pitch and pacing for every word `measured inside the frozen boundaries`
05	**modal**	one rented gpu runs it all `warm worker / four clips in parallel`

THE TEST

One hour and 44 minutes of the same audio, transcribed six times. Three times with the custom free stack on a rented gpu. Three times through the Gemini 3.5 Transcribe api, one request per clip.

WHAT CAME BACK

01 The Free Stack cost 39% less (free with Modal's $30 free credits).

But

02 The Free Stack gave the same answer every time. Gemini did not.

Three cold runs of the Free Stack returned byte-identical words, timings and speaker files. Gemini's word count changed between repeats, and its returns carried hundreds of zero-duration words and dozens of timestamps out of order.

03 One request said completed and was missing nine minutes of speech.

Nine minutes and 34 seconds out of the middle of one clip, gone, with no error returned. Same boundary in all three runs of that mode. The Free Stack found 1,595 words in that same stretch.

FULL HEAD TO HEAD

Same 103.97 minutes of real footage. Three repeat trials per configuration.

	free stack	gemini 3.5 transcribe
median processing time	**5m 57s**	4m 06s to 4m 12s
estimated cost per run	**$0.319**	$0.520
cost difference	**38.7% cheaper**	—
monthly cost to you	**$0 within modal's $30 credit**	paid api
word output	**17,494 every trial**	16,043 to 17,979
repeatability	**words and speakers byte-identical**	changed between trials
zero-duration words	**29**	516 to 649
timeline reversals	**0**	29 to 72
overlaps	**0**	46 to 88
missing section	**none found**	**9m 34.4s skipped, 3 of 3 times**
diarization	**speaker identities across the whole set**	per clip
tone and pacing	**included as prosody**	not provided
transcription api calls	**zero**	files + interactions apis

Gemini was about two minutes faster. Its words-only mode skipped the same 9m 34.4s section on all three runs while still reporting completed. On that clip it returned 750 to 767 words, against 2,394 from the Free Stack.

HAND THIS TO CLAUDE OR CODEX

Build the Free Stack to this spec. Use the exact package versions, model revisions and inference settings on this page. Do not substitute anything on the never list. When you are done, run the same audio three times and show me the checksums.

PINNED RUNTIME

environment	Debian / Python 3.11	cuda 12 / cudnn 9
audio	ffmpeg	mono pcm16 wav / 16 khz
package	faster-whisper	1.2.1
backend	CTranslate2	4.8.1
model	faster-whisper-large-v3	rev 53ecf83a5bedc5597eb8c8b34eac29e5345520ff
precision	compute_type=float32	not fp16, not int8
execution	WhisperModel(num_workers=4)	native transcribe, four clips at once
inference	vad_filter=False	word_timestamps=True
	temperature=0.0	condition_on_previous_text=False
speakers	pyannote.audio	4.0.7
model	speaker-diarization-community-1	rev 3533c8cf8e369892e6b79ff1bf80f7b0286a54ee
batching	segmentation=8 / embedding=8	one pass over the whole batch
numeric	torch 2.8.0 / numpy 2.4.6	torchaudio 2.8.0 / soundfile
access	hugging face token	community-1 weights are gated
compute	one warm NVIDIA L40S	16 cpu / 64 gib / models load once per worker

ORDER OF OPERATIONS

01	**normalize**	ffmpeg to mono 16 khz. the original file is never rewritten
02	**transcribe**	one model in memory, four clips at once, settings above
03	**freeze**	write word, start, end to disk and never edit them again
04	**assemble**	every clip into one lane, 2 s of silence between, keep the offset map
05	**diarize**	one speaker pass over that lane, then map turns back to clip time
06	**assign**	each word takes the speaker with the greatest positive overlap
07	**measure**	one prosody row per word: pause, duration, energy, voicing, pitch and slope
08	**confirm**	a human names the speakers
09	**publish**	hash the output, then write the transcript

IF THE BUILD IS RIGHT, THESE ARE TRUE

01 the same audio, run three times, gives byte-identical word and speaker files

02 every word starts before it ends, and no start runs backward

03 no two words overlap in time

04 the word count does not change between runs

05 a person listens to thirty seconds and confirms the transcript matches

NEVER

VAD on.	a vad path omitted real speech, so production runs without it
30-second chunking.	it lost words at the boundary, so whole clips are transcribed
FP16.	it produced timestamp and word-sequence variance across fresh workers
Speaker IDs per clip.	speaker 1 in clip 3 is not speaker 1 in clip 7
A model naming people.	diarization hears voices, not identities
Fuzzy text to time.	use the timestamps you froze